

Practice 1 - Setting up the Stretch Robot Simulator

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1 Introduction

The Hello Robot Stretch is a mobile manipulator for research and assistive robotics. This practice introduces the Stretch MuJoCo Digital Twin—a high-fidelity physics simulation for developing robotics algorithms.

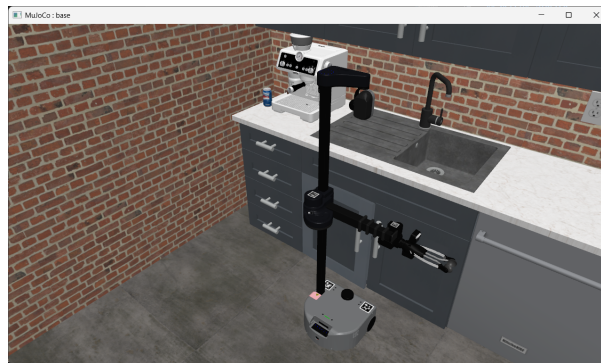


Figure 1: Stretch robot operating in the MuJoCo simulation environment

Objectives:

- Install and verify the base simulator
- Complete a block stacking challenge using teleoperation
- Optionally install and verify RoboCasa kitchen environments

2 Requirements

- Windows, macOS, or Linux laptop
- Git
- uv package manager
- Code editor (VS Code recommended)
- **Optional:** Xbox-compatible gamepad
- 5+ GB free disk space

3 Setup

Video Tutorial: <https://youtu.be/FLPB3TQcUoI>

3.1 Install uv Package Manager

Windows:

```
1 powershell -c "irm https://astral.sh/uv/install.ps1 | iex"
```

macOS/Linux:

```
1 curl -LsSf https://astral.sh/uv/install.sh | sh
```

Restart your terminal after installation.

3.2 Clone Repository

```
1 git clone https://github.com/SilentSammy/stretch_mujoco_digital_twin --
  recurse-submodules
2 cd stretch_mujoco_digital_twin
```

Linux users: If you get build errors, install:

```
1 sudo apt install build-essential python3-dev linux-headers-$(uname -r)
```

4 Part 1: Verify Base Simulation

Run the simulator:

```
1 uv run examples/teleop_demo_cam.py
```

You should see:

- MuJoCo viewer with Stretch robot
- Two colored blocks (blue box and red cylinder)

Important: Before controlling the robot, click away from the MuJoCo viewer (e.g., click the taskbar or another window). This prevents keyboard inputs from triggering simulator functions. The controls will still work when the window is unfocused.

Camera Feeds

The teleoperation demo also provides the robot's five camera feeds. They begin disabled to reduce simulator lag. Press 1 through 5 to toggle, respectively, head RGB, head depth, wrist RGB, wrist depth, and wide-angle RGB. Press Q or Escape to close the demo.

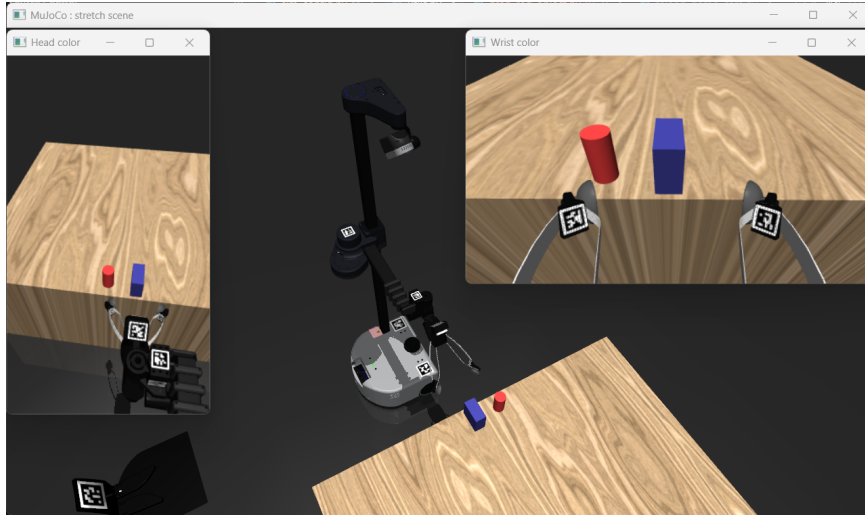


Figure 2: Head RGB, head depth, wrist RGB, wrist depth, and wide-angle camera feeds.

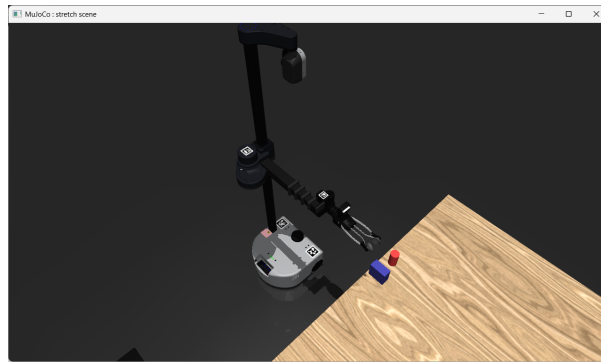


Figure 3: Default simulation environment

Press **Ctrl+C** to exit.

5 Part 2: Block Stacking Challenge

Task: Stack one block on top of the other using teleoperation.

5.1 Stretch's 9 Degrees of Freedom

1. Base Translation
2. Base Rotation
3. Lift
4. Arm
5. Wrist Yaw
6. Wrist Pitch
7. Wrist Roll
8. Gripper
9. Head Pan/Tilt

5.2 Keyboard Controls

Keys	Action
W / S	Base forward/backward
D / A	Base rotate right/left
Z / X	Lift up/down
V / C	Arm extend/retract
M / N	Gripper open/close
L / J	Wrist yaw
U / O	Wrist roll
K / I	Wrist pitch
H	Toggle head/wrist mode

Table 1: Primary keyboard mappings (see `teleop_mappings.json` for gamepad)

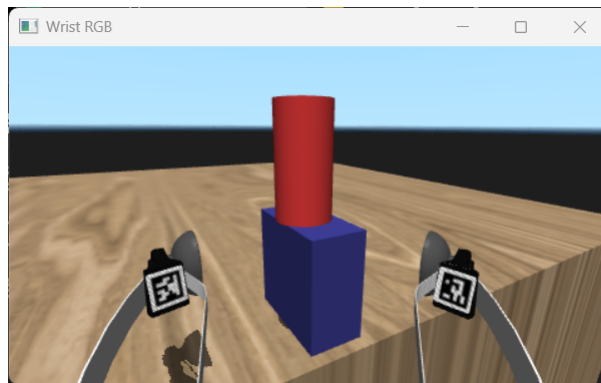


Figure 4: Target outcome: one block stacked on top of the other

Tips: Orbit the MuJoCo viewer as needed for better depth perception. Take your time.

6 Part 3 (Optional): Verify RoboCasa Environment

This optional step verifies that your computer can run the RoboCasa kitchen environments used in later practices.

Install RoboCasa dependencies:

```

1 uv pip install -e ".[robocasa]"
2 uv pip install -e "robocasa@third_party/robocasa"
3 uv pip install -e "robosuite@third_party/robosuite"
4 uv run third_party/robosuite/robosuite/scripts/setup_macros.py
5 uv run third_party/robocasa/robocasa/scripts/setup_macros.py
6 uv run third_party/robocasa/robocasa/scripts/download_kitchen_assets.py

```

Note: Asset download is 5 GB and takes some time.

In `sim_config.py`, select the RoboCasa scenario:

```

1 CONFIG = _ROBOCASA

```

Then run:

```

1 uv run examples/teleop_demo_cam.py

```

You should see a kitchen environment with cabinets, counters, and appliances.

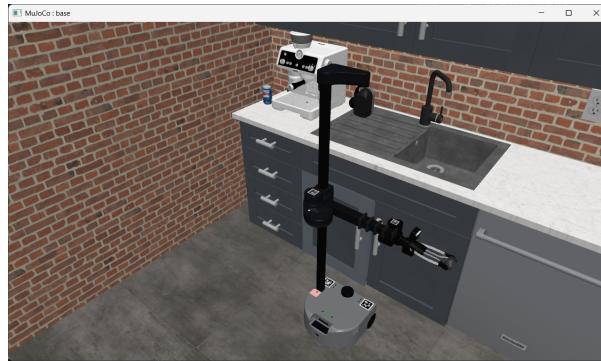


Figure 5: RoboCasa kitchen scene

7 Evidence and Submission

Submit a PDF report (1-2 pages) containing:

- **Link to a short video:** Move each joint and briefly explain what it does in your own words as it moves.
- A screenshot of the stacked blocks result (Part 2).
- **Optional:** A screenshot of the RoboCasa kitchen environment (Part 3).
- A short written description of each of the robot's 9 degrees of freedom and its corresponding control key.

File naming: LastName_FirstName_Practice1.pdf

8 Troubleshooting

Simulator won't start: Run `uv sync` then retry

Controls not responding: Click the MuJoCo viewer window to focus it

Physics issues: Restart simulation with `Ctrl+C`

8.1 Resources

- https://github.com/SilentSammy/stretch_mujoco_digital_twin
- <https://mujoco.readthedocs.io/>
- <https://hello-robot.com/stretch-3>

9 Conclusion

You've installed the Stretch simulator and completed a manipulation task. If you completed the optional section, you also verified the RoboCasa kitchen environment for future practices. Future practices will cover programmatic control, computer vision, and autonomous planning.

Good luck with the block stacking challenge!